INCH-POUND

A-A-54781
29 April 1994

COMMERCIAL ITEM DESCRIPTION

VENTILATOR, CRITICAL CARE

(Volume Cycled, Apnea Ventilation Back-Up and Volume Monitoring)

The General Services Administration has authorized the use of this Commercial Item Description.

This Commercial Item Description covers a microporcessor controlled, pneumatically operated, pressure limited, volume cycled ventilator.

Salient characteristics:

The ventilator shall be an electrically powered, microprocessor controlled, pneumatically operated, volume cycled, pressure limited critical care ventilator with back up apnea ventilation and volume monitoring for adult and child patients.

The ventilator shall be in conformance with ASTM F 1100, Ventilators Intended for Use in Critical Care, as applicable to adult and child patients, except that the manufacturer need not provide an oxygen concentration monitor nor provide a recommended source for such a monitor (at least, the manual shall recommend that oxygen concentration be monitored). The ventilator shall be listed/approved as being in conformance with NFPA 99, (Devices used in Health Care Facilities), Public Law 90-602, (Subchapter J of Title 21, Electronic Devices Generating Radiation), ANSI/AAMI ES1, (Electromedical Apparatus), and other safety requirements as applicable by an independent nationally recognized testing laboratory (i.e.: UL listed, or approved by CSA, ETL, or City of LA).

AMSC N/A FSC 6530

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The ventilator shall be electrically and pneumatically powered. It shall be operational with inputs of 120 VAC, 50/60 Hz or 11.8 - 16 VDC, 5 Amps (maximum), and 30-70 psig air and oxygen.

The following compatible accessories conforming with the national and/or international applicable controlling standard(s) which shall be listed/approved as being in conformance with applicable safety standards by an independent nationally recognized testing laboratory, shall be available for individual selection (the listed cover may be a standard component with the ventilator in lieu of being offered as an accessory):

* Blender, Air/Oxygen, With Hoses, Connectors, and Mounting Bracket for mounting on stand or compressor. The blender shall be designed to allow the ventilator to operate with flow rates of, at least, 2 - 120 Lpm.

* Humidifier, Heated, with Electric Cord and Plug, Hoses, Connectors, and Mounting Bracket for mounting on stand or compressor. The humidifier shall include a temperature sensor and a controller and be capable of delivering a constant gas flow up to, at least, 60 Lpm at 37° C. The controlling standard is ANSI Z79.9.

* Air Inlet Filter and Watertrap. Complete assembly with filter element. With bracket for mounting to Stand or Compressor if required.

* Hospital Grade Electrical Outlet Strip for attachment to stand or compressor.

* Interface Module, RS232. Process ventilator data for output to serial printer or central computer. With power cord and plug.

* Consumable Items Package. All consumable type items recommended by the manufacturer (i.e. Package of valve diaphragms, valve body(ies), filter elements, etc. as recommended). This package may be offered as separate items as opposed to a listing of items in a single package.

* Support Arm, Patient Breathing Circuit, with Circuit Clips, and Arm Bracket for mounting on stand or compressor.

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* Circuit, Patient Breathing, Reusable, Autoclavable. Complete kit for functional use.

* Pole, IV, 5/8 inch x minimum 33 inches, with bracket for mounting on stand or compressor.

* Stand, Ventilator, Mobile with Casters. Includes required bracket(s) for ventilator and attachment of other brackets for accessories.

* Cover, Front Panel Protective (may be a standard item with the ventilator as opposed to being listed as an accessory).

* Air Compressor Assembly with Electric Cord and Plug, and Bracket Assembly, Mobile, in Cabinet with Casters. 115 VAC, 60 Hz. Includes low pressure, high temperature, and power interruption alarms and auto switchover in case of primary air source low pressure or failure. Not to be used with stand, transport package, or wall mount package.

* Transport Package. Includes battery/charger pack and cable, quad yoke manifold and air/O_2 regulator(s) for two O_2 and two air size "E" cylinders, air and O_2 hoses with connectors, and brackets/supports for stand mounting with four size "E" cylinders (cylinders not included).

* Wall Mount Package. Includes brackets/supports for ventilator, and brackets/supports to attach brackets/items for O_2 blender, humidifier, and circuit support arm. Includes required extra hoses, elbows, and connectors.

The measurements of the ventilator shall not exceed 12 x 16 x 16 inches and the weight shall not exceed 40 pounds.

The microprocessor(s) shall control the overall ventilator functions including timing (such as breath rate), volume control, volume monitoring, and valve motion for gas flow to patient and exhalation. The flow and exhalation valves shall be electromechanical type.

The unit shall provide two basic selectable ventilation modes for assist/control, and synchronous intermittent mandatory ventilation (SIMV)/continuous positive airway pressure (CPAP). It shall provide selectable square and decelerating waveform for assist/control and SIMV/CPAP modes. Selection

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of assist or control submodes and SIMV or CPAP submodes may be determined by other settings that are provided involving tidal volume control, peak flow control, breath rate control, positive end expiratory pressure (PEEP/CPAP) control, and sensitivity control.

The ventilator shall support at least three (3) types of breath control, volume control to satisfy tidal volume and waveform settings, spontaneous breath using pressure control to maintain stable PEEP/CPAP setting, and pressure support using flow control to maintain the pressure support setting.

The ventilator shall have a switch to select 12 VDC or 120 VAC for external power supply.

The ventilator shall provide apnea back up ventilation (adjustable from 0 to 80 BPM) and shall have adjustable apnea interval (10 to 60 seconds).

The ventilator shall provide an inspiratory hold function and have variable alarm volume control.

It shall limit, during all modes of operation, the peak pressure delivered by adjusting the high pressure limit control (1-140 cm H_20).

The ventilator shall have stand, wall, or compressor mount capability.

The ventilator shall provide and display control settings for:

Control	Settings	Type display
Power	On/Off	LED
Mode control	Assist/Control-SIMV/CPAP	Knob position
Tidal volume	50 to 2000 ml	LED
Breath rate	0 to 80 BPM	LED
Peak flow	10 to 120 Lpm	LED
Waveform	Square or decelerating	Knob position

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Control	Settings	Type display
Pressure support	Off to 50 cmH_2O insp.	LED
Assist sensitivity	-1 to -20 cmH_2O, Off	LED
PEEP/CPAP	0 to 30 cmH_2O	LED
Manual breath	Touch button activated	Depress button
Inspiratory hold	Touch button activated	Depress button/LED display of measured pressures
Expiratory hold	Touch button activated	Depress button/LED display of measured pressures
Sigh	On/off - 1 sigh/100 br.	LED
Sigh volume	1.5 x Tidal Volume	See Tidal Vol.
Alarm silence	Button - 60 seconds	LED
Alarm reset	Button	Not required
AC/DC power source	Switch position	Switch pos.
Data linkage	RS232 port in use	N/A
Alarm volume	Plus and minus 4 to 6 dB of approximately 80 dB	Adj. knob
Backup breath rate	0 to 80 BPM	LED
Event Detection Module (EDM)	Touch button activated @ power up	LED
Fiber Optic Interface	Graphics interface	N/A

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The ventilator shall display monitored values for (simultaneous display not required):

Monitoring	Range of value	Type display
Power on	On/off	LED
Minute volume	0 to 130.0 L	LED
Tidal volume	0 to 9999.9 ml	LED
Breath rate	0 to 255 BPM	LED
I:E ratio	1:10 to 99 or flashing Inverse 1.0 to 99:1	LED
Airway pressure	-20 to +140 cmH_2O	Vacuum/ pressure gauge
Battery	Plugged in DC source	LED
Inspiratory effort	Event with no value	LED
Hour meter	0 to 99,999 hours	LED

The ventilator shall display and provide audible alarm settings according to purpose via color, flashing, and sound type (continuous, pitch, cycled beeps/tone) etc.:

Alarm	Settings	Display	Audible
High pressure limit	1 to 140 cmH_2O	LED	Yes
Low peak pressure	Off, 2 to 140 cmH_2O	LED	Not req*
Low baseline press	-20 to +30 cmH_2O	LED	Not req*
Low minute volume	0 to 99.9 1	LED	Not req*
High breath rate	3 to 150 BPM	LED	Not req*
Low inlet gas press	17 psig	LED	Yes
Apnea interval	10 to 60 seconds	LED	Yes
Ventilator inoperative	Program controlled	LED	Yes

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Alarm	Settings	Display	Audible
Breathing circuit integrity	Program controlled to allow breathing	LED	Not req*
Mode/waveform discrepancy	Program controlled per waveform/vol-time	LED	Not req*

*Not required if type of alarm is immediately discernible.

The ventilator shall have an Event Detection Module (EDM) which captures and records events that may reflect ventilator operation and/or performance.

The ventilator shall provide an expiratory hold function which allows the user to measure and display end expiratory pressure.

The ventilator shall provide a fiber optic link which allows for connection to a separate graphics monitor capable of displaying real-time waveform graphics.

Workmanship. The ventilator shall be free from defects which detract from its appearance or impair its serviceability.

Unit. Each (EA). One ventilator, as specified, constitutes one unit.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

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Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, with an AQL of 0.0 (percent defective) and an inspection level of I. The unit of product for examination purposes shall be one ventilator, as specified in this description.

Sampling for tests. Sampling for tests, including dimensional test, shall be conducted in accordance with MIL-STD-105, with an AQL of 0.0 (percent defective) and an inspection level of I. The unit of product for test purposes shall be one ventilator as specified in this description.

Tests. Tests shall be conducted to determine compliance with specification requirements. Where feasible, the same sample shall be used for the determination of two or more test characteristics. As a minimum, applicable testing addressed in ASTM F 1100 shall be performed (contractor's protocol is acceptable under the restraints of ASTM F 1100).

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard 376, and all other requirements of this document are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this Commercial Item Description, and that the product conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

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Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

Ordering data. Intermediate/exterior package quantities, labeling, and marking must be specified in the contract and/or order.

NOTES:

1. This specification requires the offeror/bidder to have available certain accessories enabling the Government to purchase the accessories separately. In addition to citing this specification, the procurement document should require the offeror/bidder to provide part number(s), description, literature, and separate price for each accessory assembly that is required to be available (see first paragraph in specifications portion of Section C of the procurement document).

2. The ordering activity should select required accessories in order to have an operating ventilator. The ordering activity should state type of mounting required (stand, wall, or compressor mount).

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3. The contracting officer/buyer must indicate the type of mounting required (see second paragraph in specifications portion of Section C of the procurement document).

4. The following National Stock Number is covered by this document.

6530-01-348-0270

5. ANSI standards are available from the American National Standards Institute Inc., 11 West 42nd Street, New York, NY 10036.

MILITARY INTERESTS:

Custodians:

Air Force - 03
Army - MD
Navy - MS

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

FDA-MPQAS
USPHS
VA - OSS

Project Number: 6530-2386

Location: VENTR12C.DOC

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	A-A-54781	940429

3. DOCUMENT TITLE
VENTILATOR, CRITICAL CARE (Volume Cycled, Apnea Ventilation Back-Up and Volume Monitoring)

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
Defense Personnel Support Center ATTN: DPSC-MSE	(1) Commercial 215-737-2870	(2) AUTOVON 444-2870
c. ADDRESS (Include Zip Code) 2800 South 20th Street Philadelphia, PA 19145	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON: 289-2340	

DD Form 1426, OCT 89 *Previous editions are obsolete.* 198-290

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